

SCIENCE SCHOLARS ACADEMY

DIFFERENTIATION

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(Lesson 2)

Lesson targets:

(1). Talk about Implicit & Parametric

Approach.

(2). $\frac{d}{dx}$ (Inverse trig functions) ✓✓✓

(3). $\frac{d}{dx}$ (exponential functions)

(4). $\frac{d}{dx}$ (Natural logarithms) + Logarithmic functions.

(5). Mixed Questions!

END

Note: for Basics of; → Check

(a). $\frac{d}{dx}$ (Implicit functions)

(b). $\frac{d}{dx}$ (Parametric functions)

Diff (1)

Revist & S.5 Term 2

(Topic 10)

→ You cannot make (y) the subject;

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IMPLICIT APPROACH:

Recall → Given that $xy^2 + x^3 = x^2y + y^2$ ✓

find $\frac{dy}{dx}$.

$$xy^2 + x^3 = x^2y + y^2$$

$$\frac{d}{dx}(xy^2 + x^3) = \frac{d}{dx}(x^2y + y^2)$$

$$\left[y^2(1) + x(2y)\frac{dy}{dx} \right] + 3x^2 = \left[y(2x) + x^2(1)\frac{dy}{dx} \right] + 2y\frac{dy}{dx}$$

$$y^2 + 2xy\frac{dy}{dx} + 3x^2 = 2xy + x^2\frac{dy}{dx} + 2y\frac{dy}{dx}$$

$$2xy\frac{dy}{dx} - x^2\frac{dy}{dx} - 2y\frac{dy}{dx} = 2xy - y^2 - 3x^2$$

$$(2xy - x^2 - 2y)\frac{dy}{dx} = 2xy - y^2 - 3x^2$$

$$\ast \frac{dy}{dx} = \frac{2xy - y^2 - 3x^2}{2xy - x^2 - 2y}$$

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For trig: Given that;
 $x \tan y - x^2 = y \sin x$ find $\frac{dy}{dx}$.

$$\frac{d}{dx}(x \tan y - x^2) = \frac{d}{dx}(y \sin x)$$

$$[(\tan y)(1) + x \sec^2 y \frac{dy}{dx}] - 2x = (\sin x)(1) \frac{dy}{dx} +$$

$$\tan y + x \sec^2 y \frac{dy}{dx} - 2x = \sin x \frac{dy}{dx} + y \cos x$$

$$x \sec^2 y \frac{dy}{dx} - \sin x \frac{dy}{dx} = y \cos x - \tan y + 2x$$

$$(x \sec^2 y - \sin x) \frac{dy}{dx} = y \cos x - \tan y + 2x$$

$$\frac{dy}{dx} = \frac{y \cos x - \tan y + 2x}{x \sec^2 y - \sin x}$$

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Brain teasers: find dy/dx

①. $y^3 + x^2 \sin x = xy - y^2 \tan x$

②. $y x^2 + xy^{xy} = \frac{x \operatorname{cosec} y}{y} \times y$

$x^2 y + xy^2 = x \operatorname{Cosec} y$

(3). $y \sin 2x = x^2 \cot\left(\frac{y}{6}\right)$

PARAMETRIC APPROACH → find dy/dx

* Recall → Parametric Algebra

$y = 1 + t^2$ $x = t^3$

Parameter $\rightarrow \frac{dy}{dt}$ $\frac{dx}{dt}$

$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} \rightarrow \frac{1}{\frac{dx}{dt}}$

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Straight to Trig:

$x = \cos \theta$ and $y = \sin \theta$. find $\frac{dy}{dx}$.

$$\frac{dx}{d\theta} = -\sin \theta$$

$$\frac{dy}{d\theta} = \cos \theta$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{d\theta} \times \frac{d\theta}{dx} \\ &= \cos \theta \left(-\frac{1}{\sin \theta} \right) \end{aligned}$$

$$\frac{dy}{dx} = -\cot \theta$$

TASK: Find

$$\frac{d^2y}{dx^2}$$

MUST FIRST VISIT DIFF. (1)

$$\begin{aligned} \frac{d^2y}{(dx)^2} &= \left[\frac{d}{d\theta} \left(\frac{dy}{dx} \right) \right] \left(\frac{d\theta}{dx} \right) \\ &= -(-\csc^2 \theta) (-\csc \theta) \end{aligned}$$

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Brain teasers:

$x = \cos 2t$ & $y = \tan^2 t$. Find $\frac{dy}{dx}$.

$$\frac{dx}{dt} = 2(-\sin 2t)$$

$$\frac{dx}{dt} = -2\sin 2t \quad \checkmark$$

$$y = (\tan t)^2$$

$$\frac{dy}{dt} = 2(\tan t)'(\sec^2 t)$$

$$\frac{dy}{dt} = 2\tan t \sec^2 t \quad \checkmark$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \left\{ \frac{1}{\frac{dx}{dt}} \right\}$$

$$\frac{dy}{dx} = (2\tan t \sec^2 t) \left(\frac{1}{-2\sin 2t} \right)$$

$$\frac{dy}{dx} = \frac{-\tan t \sec^2 t}{\sin 2t} \quad \checkmark$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \left(\frac{dt}{dx} \right) \quad \checkmark \checkmark$$

Refer to video of Parametric functions!

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Brain teaser

$$x = \cos a - \sin a, \quad y = \sin^2 a - \cos^2 a$$

find dy/dx hence $\frac{d^2y}{dx^2}$

(2). $\frac{d}{dx}$ (inverse trig functions)

Recall; $\cos^2 x + \sin^2 x = 1$
 $\cos^2 x = 1 - \sin^2 x$ ✓

OR

$$\sin^2 x = 1 - \cos^2 x$$
 ✓

①. $\cos x = \sqrt{1 - \boxed{\sin^2 x}} \rightarrow (\sin x)^2$ ✓✓

②. $\sin x = \sqrt{1 - \boxed{\cos^2 x}} \rightarrow (\cos x)^2$ ✓✓

Also;

$$1 + \boxed{\tan^2 x} = \boxed{\sec^2 x}$$
 ✓

$$\rightarrow (\tan x)^2$$

$$\rightarrow \sec x = \sqrt{1 + (\tan x)^2}$$
 ✓✓✓

$$(\tan x)^2 = (\sec x)^2 - 1$$

$$\rightarrow \tan x = \sqrt{(\sec x)^2 - 1}$$
 ✓✓✓

Also

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$$1 + \boxed{\cot^2 x} = \boxed{\operatorname{cosec}^2 x}$$

$\downarrow \quad \quad \quad \downarrow$
 $(\cot x)^2 \quad \quad \quad (\operatorname{cosec} x)^2$

$$\sqrt{1 + (\cot x)^2} = \operatorname{cosec} x$$

$$\cot^2 x = \operatorname{cosec}^2 x - 1$$

$\downarrow \quad \quad \quad \downarrow$
 $(\cot x)^2 \quad \quad \quad (\operatorname{cosec} x)^2$

$$\cot x = \sqrt{(\operatorname{cosec} x)^2 - 1}$$

Examples:

(a). $y = \sin^{-1} x$. Find $\frac{dy}{dx}$.

Recall:

$$\sin x = 0.5$$

$$\rightarrow x = \sin^{-1} 0.5$$

$$\rightarrow x = \sin^{-1} 0.5$$

$$\sin x = 0.5$$

GOING
BACK ✓

$$p = \sin^{-1} q$$

$$\sin p = q \quad \checkmark$$

$$u = \cos^{-1} v$$

$$\cos u = \checkmark$$

$$\tan^{-1} a = b$$

$$a = \tan b \quad \checkmark$$

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(a). $y = \sin^{-1} x$ Find $\frac{dy}{dx}$.

B_1 $\sin y = x$ { Becomes implicit }.

$$\cos y \frac{dy}{dx} = 1 \quad B_1 B_1$$

$$\frac{dy}{dx} = \frac{1}{\cos y} \quad B_1 \rightarrow \text{Can be eliminated}$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 - (\sin y)^2}} \quad B_1$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}} \quad A_1 \checkmark \checkmark$$

(b). $y = \cos^{-1} x$ Find dy/dx !

$$\cos y = x \quad \checkmark \checkmark \checkmark$$

$$-\sin y \frac{dy}{dx} = 1 \quad \checkmark \checkmark \checkmark$$

$$\frac{dy}{dx} = -\frac{1}{\sin y} \quad \checkmark \checkmark$$

$$\frac{dy}{dx} = -\frac{1}{\sqrt{1 - (\cos y)^2}} \quad \checkmark \checkmark$$

$$\frac{dy}{dx} = -\frac{1}{\sqrt{1 - x^2}} \quad \checkmark \checkmark$$

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[C]. If $y = \tan^{-1} x$ find dy/dx .

$$\tan y = x \checkmark$$

$$\sec^2 y \frac{dy}{dx} = 1 \checkmark$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y} \checkmark$$

$$\frac{dy}{dx} = \frac{1}{1 + (\tan y)^2} \checkmark$$

$$\frac{dy}{dx} = \frac{1}{1 + x^2} \checkmark$$

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More complex ones: Find dy/dx .

(a). $y = \sin^{-1}\left(\frac{x}{3}\right)$

$$\sin y = \frac{x}{3} \checkmark$$

$$\cos y \frac{dy}{dx} = \frac{1}{3} \checkmark$$

$$\frac{dy}{dx} = \frac{1}{3 \cos y}$$

But $\cos y = \sqrt{1 - \sin^2 y}$

$$\cos y = \sqrt{1 - \left(\frac{x}{3}\right)^2}$$

$$= \sqrt{1 - \frac{x^2}{9}}$$

$$= \sqrt{\frac{9 - x^2}{9}}$$

$$\cos y = \frac{1}{3} \sqrt{9 - x^2}$$

$$\frac{dy}{dx} = \frac{1}{3 \left(\frac{1}{3} \sqrt{9 - x^2} \right)}$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{9 - x^2}} \checkmark$$

(b). $y = \cos^{-1}\left(\frac{x}{5}\right)$

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⑥. $y = \cos^{-1}\left(\frac{x}{5}\right)$

$\cos y = x/5$ ✓

$-\sin y \frac{dy}{dx} = \frac{1}{5}$ ✓

$\frac{dy}{dx} = \frac{-1}{5 \sin y}$ ✓

$\sin y = \sqrt{1 - (x/5)^2}$
 $= \sqrt{1 - x^2/25}$ ✓
 $= \sqrt{\frac{25 - x^2}{25}}$

$\sin y = \frac{1}{5} \sqrt{25 - x^2}$

Consider; $\sin y = \sqrt{1 - (\cos y)^2}$
 $\rightarrow \frac{dy}{dx} = - \frac{1}{5 \left(\frac{1}{5} \sqrt{25 - x^2} \right)}$ ✓

$\frac{dy}{dx} = \frac{-1}{\sqrt{25 - x^2}}$ ✓

Brain teasers: Find dy/dx

(a). $y = \sin^{-1} x^2$ (b). $y = \cos^{-1}(x^2 - 3)$

(c). $y = \cot^{-1} x$ (d). $y = \tan^{-1}\left(\frac{x}{1-x^2}\right)$

(e). $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ (f). $y = \sin^{-1}\left(\frac{3+5\cos x}{5+3\cos x}\right)$

Research: find $\left[\frac{dQ}{dP}\right]$ for $Q = \sin^{-1} 5P^3$

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Imeg ✓
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Ug!

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